The Science Behind CPTG

A Framework of Scales for Gravity, Structure, Cosmology, and Nuclear Reactions

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Abstract

Curvature Polarization Transport Gravity (CPTG) explores an alternative way of understanding the gravitational behavior of the Universe: ordinary matter remains its physical source, while the geometry associated with that matter may develop an additional organized response. The theory retains General Relativity and adds curvature polarization and transport, recovering the familiar Einstein description when those additional contributions become negligible. Its central question is whether this common geometric architecture can explain phenomena that are usually assigned separate physical remedies. Polarization and transport describe the proposed gravitational geometry itself, not an additional material component.

This article introduces CPTG without requiring advanced mathematics. Galaxy rotation curves show how the framework reconstructs measured gravitational fields; galaxy clusters test the same gravitational architecture through the pressure of hot gas; and Structural Mode N measures the organization of a field after it has been solved, revealing information that baryonic matter, overall shape, and size alone do not determine. In black holes, CPTG proposes a finite interior joined to the successful General Relativistic exterior, with a distinction between localized source mass and the geometry's contribution to the exterior gravitational state. Its compact solutions also provide a way to calculate gravitational time dilation at a location identified by Structural Mode N.

At cosmological scales, CPTG describes one native geometric history with different acoustic and luminosity observational projections. Its redshift–phase relation applies irrespective of distance, while independent measurements of distance and motion retain their own observational roles. CMB comparison maps, the CMB-to-DESI cosmic standard ruler, and the Hubble tension provide examples of how these projections are examined. In primordial nuclear physics, related geometric organization is implemented in reaction networks without replacing nuclear dynamics. Throughout, numerical reconstructions, independently withheld tests, and proposed physical interpretations are distinguished, so that the scope of each result remains clear.

1 A different question about gravity

The stars in a galaxy move under gravity. Hot gas in a cluster of galaxies is held in place by gravity. And near a black hole, gravity can affect both the motion of matter and the passage of time. These are familiar consequences of gravitation, but they are normally studied with very different kinds of evidence. The question linking them is simple: could the gravitational response depend not only on how much ordinary matter is present, but also on how the surrounding geometry is organized?

Curvature Polarization Transport Gravity, or CPTG, investigates that possibility. In conventional General Relativity, matter and energy determine spacetime curvature, and curvature guides motion. CPTG keeps this relationship. It proposes that, in certain physical states, the geometry has an additional response described by *curvature polarization* and *curvature transport*. These terms refer to organization within the gravitational field, not to invisible material added around a galaxy. Where their contribution becomes unimportant, the theory is constructed to return smoothly to Einstein gravity [1].

The motivation is not confined to a single astronomical puzzle. Galaxy rotation curves, the gravitational support of galaxy clusters, extreme compact objects, differences between cosmological measurements, and the primordial lithium discrepancy all provide opportunities to investigate whether a common organizing principle has been overlooked. CPTG does not claim that one simple equation calculates all these phenomena. Its gravitational field equations have different reductions for different geometries; its nuclear construction uses nuclear reactions rather than a gravitational force inserted into an atomic nucleus.

A useful principle runs throughout the research: a calculation should specify its inputs, its equations, and the observables to be compared before its result is judged. An agreement is more informative when it survives a new dataset, a different reconstruction, or a measurement that was not used to produce the result. This distinction matters particularly in the examples below.

2 How ordinary matter and geometry work together

It helps to separate three ideas that can otherwise become confused. *Matter* is the physical source: the stars, gas, and other ordinary material in an object. *Polarization* describes the way the geometric response is organized by the strength, gradients, and structure of the matter-supported field. *Transport* describes how this organized response is carried or redistributed through a system. In an extended galaxy, for example, it matters how the gravitational support changes from one radius to another, not just what happens at one point.

These words should not be read as a claim that space contains an extra fluid of material particles. CPTG puts the additional contribution on the geometric side of its gravitational equation. Its parent equation retains Einstein curvature and ordinary stress–energy, together with added geometric polarization and transport terms that must satisfy the appropriate conservation requirements [1]. This is why the framework can accommodate

familiar Solar System and black-hole exterior behavior in the limit where its extra geometric response fades.

Nor does the theory require gravity always to become stronger. Different physical arrangements can permit different kinds of response, including weak-response systems. This is an important difference between measuring an object's gravitational state and assuming that every object must receive the same amount of extra support. An object must be evaluated according to its actual source, geometry, motion, and physical boundary conditions.

CPTG is therefore best understood as an attempt to uncover a common *rule of organization*. The same gravitational architecture can be reduced for rotating disks, approximately spherical clusters, and compact spacetime. The details of each reduction are important to researchers, but the everyday physical message is straightforward: the arrangement of a system can matter as well as its total mass.

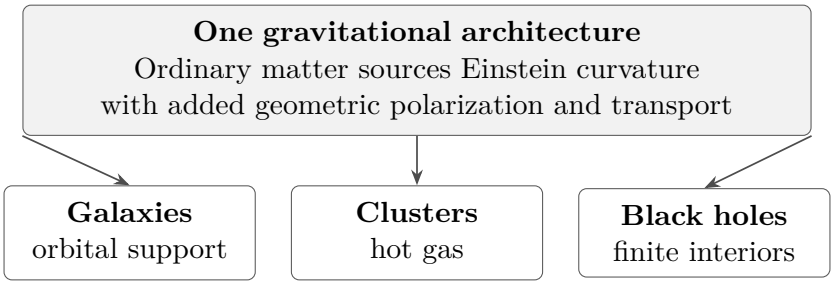


Figure 1: A conceptual map, not an observational fit. Galaxies, pressure-supported clusters, and black holes require different calculations, but CPTG relates them to a shared gravitational foundation.

3 How ΛCDM, MOND, and CPTG differ

Several approaches seek to explain why the motions of stars, gas, and galaxies sometimes require more gravitational support than visible matter seems able to provide under a simple Newtonian estimate. They begin from different physical ideas, and those differences matter more than the acronyms.

Λ*CDM* (pronounced "Lambda-CDM") is the prevailing cosmological framework. The letter Λ denotes the cosmological constant, ordinarily associated with dark energy; CDM means *cold dark matter*. It uses General Relativity together with an inferred, non-baryonic matter component and a cosmological constant. With these ingredients, it reproduces many observed features of the cosmic microwave background and large-scale cosmic structure [33]. Those agreements are important, but they do not establish the physical identity of the proposed dark component, which has not been directly detected. At the scale of individual galaxies, the inferred halo mass and distribution can depend strongly on the observational data and on assumptions used to interpret them.

Modified Newtonian Dynamics (MOND) begins with a different proposal: the effective rule connecting matter to motion changes in very low-acceleration environments. Its characteristic acceleration scale, a_0, captures important regularities in galaxy rotation. MOND began as a modification of dynamics; several relativistic theories have subsequently

been developed around its central idea. It should not be mistaken for a dark-matter particle model [8, 9].

CPTG makes a third proposal. It retains Einstein's gravitational curvature and the ordinary matter source, while adding geometric curvature polarization and transport. Its additional gravitational response is determined by the solved structure of the system, not by placing an invisible halo in the material source and not by applying a single universal low-acceleration correction. *CPTG is not a dark-matter model, and polarization and transport are not perturbations of a dark-matter field.* Its native cosmology likewise does not introduce separate cold-dark-matter or dark-energy constituents [1, 2, 17].

Approach	**Central physical idea**	**How it treats gravitational discrepancies**
ΛCDM	General Relativity with cold dark matter and a cosmological constant.	An unseen, non-baryonic component supplies gravity; Λ describes cosmic acceleration.
MOND	Dynamics changes below a characteristic acceleration scale.	A low-acceleration rule modifies how motion reflects ordinary matter.
CPTG	General Relativity with curvature polarization and geometric transport.	The solved field gains geometric response from polarization and transport, without adding a material halo.

Table 1: The physical distinction among three research approaches. This is a comparison of governing ideas, not a ranking or a statistical performance test.

Observations must decide. CPTG takes a different kind of risk: once its equations, source data, boundary conditions, and numerical procedure are declared for a test, an unfavorable result cannot be rescued by assigning the object an otherwise unseen matter halo or silently changing the prescription afterward. If better observations become available, the calculation can be repeated with those new inputs, but the original result and the reason for the change remain part of the record. CPTG must succeed or fail against the observations under those stated conditions; measured-field reconstructions must still be distinguished from genuine predictions of withheld measurements.

The approaches therefore make different physical commitments. ΛCDM models additional material components, MOND changes the low-acceleration dynamical relation, and CPTG adds a specified geometric response to Einstein gravity. Each must be judged by its actual observational tests, not merely by its ability to restate what has already been measured.

4 Galaxies: learning from the way they rotate

A spiral galaxy is a natural place to measure gravity. Astronomers track the orbital speeds of gas and stars at different distances from the center, producing what is called a *rotation curve*. The outer regions of many galaxies move faster than a simple calculation from the observed ordinary matter and conventional Newtonian approximation would suggest. In the standard ΛCDM picture, a non-baryonic dark-matter halo supplies the additional gravitational support. CPTG instead examines whether the geometry associated with the observed galaxy can account for the measured support without placing such a halo in the material source.

To investigate a particular galaxy, astronomers first measure its rotation and estimate the

distribution of stars and gas. CPTG then asks what organized geometric response can support that observed state. Its local structural acceleration, $a_\star$, is determined separately at each measured radius, allowing polarization and transport to respond to the changing structure of the galaxy [3, 2].

The resulting calculation is a *measured-field reconstruction*: it uses the observed motion as an input, rather than predicting a previously unknown rotation curve from the stars and gas alone. This distinction does not diminish the value of testing a common law across many galaxies; it tells us what the agreement establishes and what a more independent future prediction would require.

The SPARC database, a widely used collection of galaxy rotation curves and stellar/gas mass models, makes it possible to study many galaxies using the same observational structure [10]. In the published 153-galaxy primary-sample study, the resolved calculation covered 3,168 measured radii and reported acceleration-space residual scatter of about 0.10 dex [3]. That scatter describes how closely reconstructed and measured accelerations agree on a logarithmic scale; it is not a blind prediction from the ordinary-matter distribution alone. The technical paper reports the full velocity-fit statistics and uncertainty assumptions. Results from a different numerical implementation should not be substituted without identifying that change.

The significance of a catalog-scale test is its variety: the same framework must contend with bright and faint systems, changing gas fractions, and very different observed rotation patterns. CPTG also examines galaxies with unusually weak additional response, a valuable counterpoint to objects requiring substantial geometric support [2]. A reader need not follow the numerical solver to understand what is being tested: can the field equations remain coherent across many different observed systems without supplying a new unseen mass distribution for each one?

5 Galaxy clusters: a second way to weigh gravity

Galaxies do not only rotate in disks. A galaxy cluster contains many galaxies and a large amount of hot, diffuse gas. That gas emits X-rays and is supported partly by its own pressure. Where the gas is sufficiently close to pressure balance, its temperature and density distribution can be used to infer the gravitational field needed to hold it in the cluster. That inference depends on the pressure-support assumptions and is a very different measurement from the rotation of a spiral galaxy.

In the CPTG seven-cluster program, the ordinary matter source is constructed from observed gas and stars, while the supporting field is reconstructed from the pressure of the hot gas. The XMM-Newton Cluster Outskirts Project (X-COP) provides measured temperature, density, and pressure profiles for this work [5, 6]. In Abell 2029, for example, the cluster's X-ray and pressure measurements are used to infer the gravity required to support its gas, and CPTG reconstructs that field over a declared radial range. The same polarization–transport law used for galaxies is evaluated in the cluster's spherical geometry. Across seven clusters and 294 retained radial points, the primary X-COP analysis reports an aggregate reduced χ^2 of about 0.975 over its selected support domains [2].

The analysis also applies a separate pressure-support reconstruction using the Archive of Chandra Cluster Entropy Profile Tables (ACCEPT), derived from Chandra observations [7]. This check explores how the reconstructed gravitational field responds to a different observational treatment of the hot gas. X-COP and ACCEPT therefore remain separate reconstructions, rather than being combined into one artificial cluster measurement [2].

Cluster measurements have real complications. Nonthermal pressure, departures from spherical symmetry, and mergers can change the meaning of a simple pressure balance; A2319, for example, requires caution because of its dynamical state. The CPTG calculations apply to declared radial domains and support reconstructions. They should not be advertised as unqualified predictions of all cluster properties. The important advance is that rotation-supported galaxies and pressure-supported clusters can now be studied under the same gravitational architecture rather than independent, unrelated equations.

6 Structural Mode N: a fingerprint of organization

Knowing an object's baryonic mass, overall shape, or physical size does not, by itself, tell us how its gravitational field is organized. Two objects that look similar in such broad descriptions can have different measured motions or pressure support and correspondingly different solved fields. Likewise, two black holes with comparable exterior gravitational properties can have different structural states in CPTG's proposed finite-interior description. The question is how to characterize the *organization* of a completed gravitational field rather than infer it from a few visible properties alone.

CPTG defines a quantity called *Structural Mode* N. It compares the extent of a solved system with a characteristic organization length extracted from the field:

$$N = \frac{R}{\lambda}. \tag{1}$$

Here R is the declared extent of the solved structure and λ is a geometry-appropriate measure of where the field's important structural changes are concentrated. The underlying mathematical procedures are different for a galaxy, a cluster, and a rotating compact object; the common idea is a dimensionless structural fingerprint [4].

The order of calculation matters. The observed baryonic source and coarse descriptors such as shape and size are not sufficient to calculate an object's Structural Mode in advance. The gravitational law must first be applied to the observed field constraints and appropriate boundary conditions; only after the solution exists is N extracted. It is not used to choose that solution and is not a knob for making a galaxy rotate correctly. In this sense, the distinct N values revealed by solved objects are evidence that gravitational organization cannot simply be read off from baryonic matter, overall shape, or size. This is a limitation of those inputs as a complete description, not a proof that a richer set of independent data could never forecast the field. When a cluster is reconstructed by separate observational methods, each reconstruction has its own N; these fingerprints must be reported individually rather than averaged into a fictional third state.

For galaxies, a downstream classification known as the Curvature Structural Morphology

Index, or CSMI, explores whether the organization measured from the field corresponds to recognizable populations of galaxies. It is distinct from Structural Mode itself. A conventional visual galaxy type does not enter the calculation of N, although observed galaxy properties can be compared with it afterward.

The cross-regime extension makes Structural Mode more interesting than a new label for galaxy shapes. In CPTG's compact-object realization, the completed solution gives N and identifies an *organization shell* inside the proposed finite geometry. At this location CPTG can calculate *gravitational time dilation*: how much time a local clock would accumulate compared with a clock far from the black hole. The shell is selected by the measured organization of the solved field; the local clock rate is then obtained separately from the geometry's time-dilation field. Thus N does not by itself determine how fast time passes, but it tells us where to evaluate that rate. The same compact construction also provides a geometric rotation readout [4, 11].

These are specific physical quantities calculated from each completed compact state, not adjustable values assigned from a black hole's apparent size or baryonic mass. Nor does an observed X-ray oscillation automatically measure the clock at the organization shell. An observational test requires a separately established relationship between the measured signal and the calculated geometry.

7 Black holes: mass, geometry, and time

A black hole is an extreme environment in which gravity shapes light, orbital motion, and the rate at which clocks run. General Relativity has been extraordinarily successful at describing observable black-hole exteriors, including the broad behavior of matter near them. CPTG explicitly retains that successful exterior description. Its proposed change concerns what happens when the inward continuation becomes so extreme that the classical solution would approach a curvature singularity.

In the registered CPTG compact construction, the interior is required to remain finite and to join continuously onto the appropriate Schwarzschild or Kerr exterior. The familiar horizon and exterior orbital dynamics are not discarded. CPTG instead asks whether an organized polarization–transport response can support a finite strong-field state beneath the exterior description [11, 1]. This is a theoretical construction with documented overlap conditions, not an observation of an interior surface.

A second difference concerns what a black hole's mass means. Astronomers observe how the black hole affects its surroundings; those motions and light signals reveal its *exterior gravitational mass*. CPTG proposes that the same exterior effect can arise from a smaller localized physical-source mass together with an organized geometric contribution. In that picture, a black hole can appear gravitationally more massive than the material concentrated in its compact source alone. The source–geometry division is a consequence of CPTG's registered compact construction, not a direct observational weighing of matter hidden inside the horizon [11].

Spin adds another layer. The registered compact polarization relation ties together the

exterior mass, a spin-dependent geometric contribution, and a corresponding compact-source state. Once that state is solved, Structural Mode identifies the organization shell, and CPTG calculates the local passage of time and geometric rotation there. The most useful observations are those in which different measurements constrain different parts of the calculation.

The time-dilation calculation gives a concrete meaning to the phrase *structure of time*. Imagine comparing a clock near the solved interior organization shell with a reference clock far outside the black hole. The local clock generally accumulates less proper time over the same exterior interval. For the adopted conditional mass–spin branches of the published M33 X-7 and LMC X-3 compact benchmark solutions, an interval of 24 hours on the source-local, distant reference clock corresponds to approximately 19.59 hours and 13.17 hours, respectively, at their distinct organization shells [11, 4]. CPTG also calculates how the clock rate changes as the inner trapping surface is approached. These figures are model-derived time-dilation measurements of the solved geometries, not direct readings from clocks observed inside black holes.

The GWTC-5 compact analysis applies the same post-solution clock readout across its declared sample of black-hole components, alongside Structural Mode and geometric frame dragging [16]. No event-specific clock adjustment is inserted to force a desired outcome.

A particularly intriguing example is 3XMM J215022.4−055108, an X-ray transient associated with a candidate intermediate-mass black hole. Astronomers detected an X-ray brightness pattern repeating approximately every 85 seconds [12]. Such a signal may provide information about motion in the strong gravitational field close to the black hole, although its physical interpretation must be tested.

In a retrospective CPTG study, X-ray spectra from other observing epochs first establish an inner-disk scale without using the observation containing the oscillation. The 85-second timing measurement is then introduced separately. If that oscillation represents Lense–Thirring precession at the innermost stable orbit, the spectral scale and timing measurement together constrain the exterior mass and spin. A separately reported spectral mass provides a further comparison, and CPTG's fixed compact relation supplies the proposed finite-state readout [13]. This is conditional spectral–timing closure, not a prediction of the period from the spectrum alone or a direct observation of an internal CPTG clock.

Gravitational waves provide a complementary window. The international LIGO–Virgo–KAGRA collaboration has published the GWTC-5 catalog and posterior samples for observed compact-object mergers [14, 15]. CPTG's GWTC-5 study considers a declared 259-event black-hole-binary subset of the catalog, analyzes released mass–spin probability distributions in its native geometric coordinates, and reports individual structural readouts [16]. A future test directly against detector strain with native CPTG waveforms would probe a different, stronger question. The present result is a population-level coordinate analysis, not that detector-level waveform test.

Black holes consequently offer several promising lines of inquiry: finite interior structure, the relation between source and exterior gravitational mass, and calculated gravitational time dilation and geometric rotation at the Structural-Mode organization shell. Future

comparisons with directly observed gravitational-wave signals could test how that geometry manifests in the data. Questions about information and evaporation are also natural subjects for further research, but should not be presented as solved consequences until their detailed dynamics have been worked out.

8 Redshift, cosmic history, and different ways of measuring it

Redshift is the change in the wavelength of light measured by an observer. In cosmology, it is a way of identifying a stage in the history of the Universe. CPTG uses a single native geometric history, often called its π-branch. A redshift selects the corresponding native phase through a simple relationship:

$$a = \frac{1}{1+z}. \tag{2}$$

The variable a is a scale or phase coordinate and z is redshift. In this native relationship, *distance is not an input*. The same redshift–phase equation applies whether the target galaxy is nearby or very distant [1, 17]. Obtaining a distance from that phase is a separate calculation, with its own observational interpretation.

An observed line-of-sight redshift may also contain a contribution from motion relative to the chosen reference frame: a galaxy's peculiar motion, the observer's motion, or the location within a rotating object where a spectrum was taken. Accounting for these effects is part of handling the measurement correctly. It does not change the distance-independent character of the native redshift equation. Nor should an independently calibrated distance—for example from a geometric or stellar standard candle—be replaced automatically by a redshift-derived distance calculated under a different model.

Why does CPTG speak of different *projections* if there is only one native history? Astronomers use different observational methods to describe cosmic scales. Some measurements rely on patterns in the early plasma, such as the cosmic microwave background and acoustic oscillations. Others use the observed brightness of objects to infer luminosity distances. CPTG constructs acoustic and luminosity comparison coordinates from the same native geometric phase. These are two ways of reading one underlying history, not two expanding universes [17].

Consider a lightning strike. One electrical discharge produces both a flash of light and a clap of thunder. The flash and the thunder are distinct ways of observing the same event, not evidence of two separate strikes. In this analogy, CPTG's one native cosmological history is the discharge, while its luminosity and acoustic projections are two different readouts of that history. The analogy concerns their common origin; CPTG's projections are not literally light and sound traveling at different speeds, and their differing cosmic coordinates do not arise from a flash–thunder delay. Their relationships are defined by the geometry of the theory.

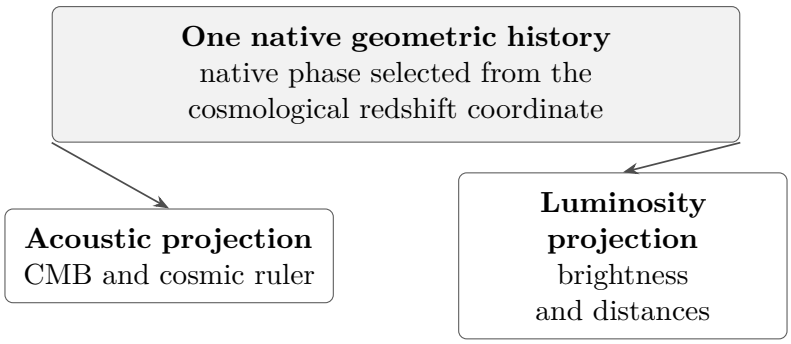


Figure 2: One native history and two observational readouts, analogous to a lightning discharge observed as a flash and thunder. The analogy illustrates a shared origin, not a signal-speed delay: the acoustic and luminosity coordinates are related by CPTG's defined projections.

This distinction also matters for high-redshift galaxies. ZF-COS-20115 is a massive, compact galaxy already quiescent at redshift about 3.7. Its observed spectrum, stellar velocity dispersion, and angular structure provide important observational constraints [18, 19]. The measured redshift selects the native CPTG cosmological phase, while an angular size must be translated through the registered CPTG distance projection before it can be expressed as a physical radius. A physical effective radius quoted in kiloparsecs is therefore not a directly observed, cosmology-independent input. The CPTG study uses these observational quantities and its declared coordinate construction in a pressure-support structural reconstruction, which can be compared with the stellar dynamics [20]. This tests a specified gravitational interpretation; it does not replace the astrophysics of star formation, mergers, or quenching.

9 The Hubble tension and the history of cosmic time

The Hubble constant summarizes the present-day expansion rate inferred within a cosmological description. Measurements tied to the early Universe and measurements constructed from local luminosity distances produce values that do not coincide as closely as expected in the standard interpretation. This disagreement is known as the *Hubble tension* [33, 34].

CPTG investigates whether the discrepancy reflects a difference between observational projections of the same native history. That native history is not obtained by assuming the conventional Friedmann–Lemaître–Robertson–Walker (FLRW) expansion equations as its governing dynamics; familiar cosmological parameters are used for observational comparison, not silently adopted as native CPTG inputs [1, 17]. Its acoustic comparison rate lies near the CMB-inferred lower value. A luminosity-side rate belongs to a different projection of that same phase history. The approach does not average the two published measurements into one number, and it does not identify a conventional comparison coordinate with an independent measurement of CPTG's native rate [21, 17].

The distinction between derived and conditioned quantities matters. The acoustic projection is supplied by the native geometric construction. Some luminosity-side comparisons have used an externally measured distance-ladder normalization, so they do not by themselves demonstrate an entirely uncalibrated luminosity-scale prediction. The achievement being examined is whether one geometrical history can support a consistent connec-

tion among these observational coordinates. A complete independent calibration would strengthen that test.

CPTG also revisits the classic *horizon problem*: why very widely separated parts of the cosmic microwave background have nearly the same temperature. Its proposed answer uses common native geometric origin and a native-to-acoustic containment calculation rather than faster-than-light signals. The published background test is anchored to the observed decoupling phase and deals with the geometry of the common background; it is not an independent theory of every tiny CMB fluctuation [22]. The cosmological account therefore links questions about cosmic distances, elapsed time, and background uniformity while preserving their distinct observational requirements.

10 The cosmic microwave background and the standard ruler

The cosmic microwave background (CMB) is the faint radiation that reaches us from the early Universe. Its almost uniform temperature and tiny fluctuations carry information about the organization of matter and radiation. Astronomers analyze those fluctuations by separating the sky into angular patterns of different sizes. The resulting measurements place unusually strong demands on any cosmological model.

One CPTG study tests a *comparison map*, using the publicly measured temperature maps from Planck and WMAP [32, 31]. A fixed CPTG angular-power envelope is placed into the harmonic coordinate supplied by those observations. The primary Planck comparison reports a correlation of approximately 0.9834 with the observed field under its declared evaluation [23]. The map tests how the chosen envelope behaves in measured sky coordinates; the observed harmonic phases and small-scale fluctuations are supplied by the observations, not independently generated by CPTG.

A different test uses the cosmic *standard ruler*. Sound waves in the early cosmic plasma helped leave a characteristic separation scale among galaxies. The Dark Energy Spectroscopic Instrument (DESI) measures that pattern across different epochs, allowing researchers to compare distances with a common acoustic scale [24, 25]. CPTG asks whether one ruler anchored in the CMB can be carried through the native geometric history and remain consistent with these measurements.

The published CMB-to-DESI test compares 13 observables: 11 lower-redshift DESI DR2 BAO measurements and two higher-redshift Lyman-α forest measurements that combine the Alcock–Paczyński and BAO information. Its covariance is assembled at the analysis level from the released lower-redshift likelihood information and a separate two-observable high-redshift block; it is not one DESI-published 13-by-13 covariance matrix [24, 25, 26]. With the CMB-anchored normalization fixed and no DESI-specific ruler adjustment, CPTG reports $\chi^2 = 13.6025$ for 13 degrees of freedom and $p = 0.4024$ [26]. This is an encouraging consistency test of the *comparison coordinates*, not an independent microscopic calculation of the ruler's physical length.

Taken together, these studies ask two powerful but distinct questions: can CPTG's declared angular organization be compared consistently with actual CMB maps, and

can its acoustic geometry carry a common ruler across cosmic history? They are not interchangeable tests, which is precisely why agreement in both settings is interesting.

11 Nuclear reactions: an organizing principle at another scale

The earliest light elements are fossils of the hot young Universe. Their surviving abundances tell us which nuclear reactions occurred and how long the reactions had to operate. Deuterium and helium give particularly demanding checks of primordial reaction calculations; lithium remains a notable puzzle. Numerical tools such as PRIMAT follow the changing temperature, nuclear reaction rates, and conserved particle inventories to calculate those abundances [28, 30]. None of these nuclear processes can be replaced by applying the galaxy gravitational equation inside an atom.

CPTG's nuclear program therefore follows the same source–polarization–transport *organizing principle* while using a separate conserved nuclear state-space formulation. Its four-sector backbone begins with free protons and neutrons (*Vertex*), moves to deuterium (*Bridge*), includes the related mass-three nuclei tritium and helium-3 (*Closure*), and concludes at helium-4 (*Saturation*). These names describe how the nuclear state is organized; they are not an assertion that every real reaction passes through one literal four-step chain [27].

The evolving reaction network still determines what happens over time. CPTG supplies a registered source geometry for how particular reactions enter that network, while the nuclear code propagates the interacting reactions to final abundances. The four-sector study reports 48 of 48 primary source/Jacobian checks passing in PRIMAT under its specified numerical criteria; additional network implementations provide further cross-checks [27, 28].

The importance of these checks is methodological: a proposed common geometric source law is being carried into several independent implementations without replacing their reaction rates and solvers. It is not a claim that gravity is the binding force inside nuclei, or that an entire table of heavy-element primordial abundances has already been derived from this construction.

12 The lithium question

Primordial lithium has long been a puzzle. Calculations of the early nuclear reaction network reproduce much of the observed light-element picture, especially the successful deuterium and helium comparisons, yet the abundance of lithium inferred from old stars is markedly lower than the standard predicted level [30]. The discrepancy asks for a correction that is selective: changing lithium should not destroy the parts of the network that already work.

CPTG investigates a fixed transport rule affecting mass seven, the group containing lithium-7 and beryllium-7. Rather than changing every nuclear reaction rate, it introduces

a declared geometric transport action within the *live evolving network*. The mass-seven survival factor associated with the isolated rule is $1/\pi$; the coupled network must still be run to determine the resulting abundances [29, 27].

In the reported PRIMAT implementation, the mass-seven transport calculation passes the preregistered numerical and abundance gates, with deuterium and helium controls retained in the specified tests. The result is a concrete reaction-network calculation, not merely a number chosen to lower a final lithium abundance. Its interpretation nevertheless depends on the declared transport law, the observational abundance comparison, and independent validation of the broader nuclear physics.

This makes lithium a particularly clear example of the CPTG research strategy: attempt one organized change to a discrepant channel, propagate it through the native physical calculation, and verify that the other measured channels remain consistent.

13 What the results demonstrate and what comes next

The strongest way to evaluate a framework is to ask exactly what each calculation has established. CPTG's research now spans much more than one kind of measurement, but not every agreement represents the same level of independence.

Application	**What the evidence establishes**
Galaxy rotation curves	Measured-field reconstruction using observed rotation and ordinary-matter profiles.
Galaxy clusters	Pressure-support reconstruction within declared observational and geometric assumptions.
Structural Mode N	A post-solution fingerprint not specified by baryonic matter, overall shape, or size alone.
3XMM compact candidate	Conditional spectral–timing closure with timing withheld from the initial spectral construction.
GWTC-5 compact population	Posterior-based compact-state and structural readouts for a declared 259-event subset.
CMB comparison maps	Observed-phase angular comparison-map closure; harmonic phases are supplied by the data.
DESI cosmic ruler	Fixed-projection observational comparison to 13 BAO/AP observables with analysis-assembled covariance.
Light-element reaction geometry	Source/Jacobian and multi-network numerical checks of the four-sector construction.
Lithium-7 transport	A specified live reaction-network transport test with deuterium and helium controls.

Table 2: Distinct CPTG applications and the evidence each supplies. Structural Mode is measured from a solved field; it is not predicted from baryonic mass, shape, or size alone.

Several lines of research naturally follow. New galaxy observations can provide velocities withheld from field construction; independent cluster measurements can test sensitivity to pressure and geometry assumptions. Better spectra and timing data for individual, well-observed black holes can sharpen the relationship among their exterior masses, spins, and proposed finite interior states. Direct detector-level waveform calculations would allow a stronger gravitational-wave test than the transformation of existing posterior samples.

Cosmology can test redshift, distance, and ruler observations without importing external-model coordinates. Independent CMB measurements and a first-principles luminosity projection remain important goals. Independent nuclear codes can test the source and transport construction.

Each new dataset offers a chance to challenge the same governing relationships in a different physical setting. The governing equations and declared test procedure must remain visible as measurements improve; a new reconstruction cannot erase an earlier result. The next advances will be most persuasive when calculations made from one set of observations continue to agree with measurements not used to construct them.

14 Conclusion

CPTG begins with a simple but far-reaching idea: the gravitational influence of ordinary matter may include an organized geometric response that depends on the system's structure. It preserves Einstein gravity while asking how polarization and transport can describe additional support in galaxies, the pressure-balanced fields of clusters, and finite compact states. Its central distinction is the additional response of geometry itself. Structural Mode N reveals a further lesson: baryonic matter, overall shape, and size do not alone specify a completed gravitational organization. The structural fingerprint is measured from the solved field, not inserted beforehand; in the compact realization it also locates where CPTG can calculate gravitational time dilation.

Beyond individual gravitating objects, CPTG uses one native cosmological history and distinct observational projections to study redshift, cosmic time, the Hubble tension, the microwave background, and the standard ruler. At nuclear scale, its corresponding geometric organization enters a conserved reaction-network construction rather than replacing nuclear interactions. These are connected research programs, not a claim that every physical regime is governed by one identical reduced equation.

The scientific value of this framework will be established by increasingly independent tests: measurements withheld from reconstruction, repeatable calculations across software and datasets, and observations capable of distinguishing CPTG from alternative explanations. The purpose of this introduction is to make those ideas and their present evidence understandable without requiring the reader to master the mathematics used to investigate them.

Data, code, and further reading

This paper is an accessible account of the research literature and does not contain a new numerical fit. Mathematical derivations, data selection, input dependencies, uncertainty treatment, numerical checks, and the precise limits of each result are supplied in the cited technical papers. Supporting code is available in the CPTG repository, including a versioned SPARC workbench release; results remain specific to the cited methods.

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